Juan Camilo Melo López

Cloud & DevOps Engineer | AWS, Kubernetes, Terraform & GitOps
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Summary

Cloud and DevOps engineer with over a year of hands-on project experience building infrastructure: Terraform on AWS through OIDC pipelines, GitOps with ArgoCD on Amazon EKS, security-gated CI/CD, Kubernetes operations and Prometheus/Grafana observability. Also designs cloud architecture on Azure, backed by a strong networking foundation (MikroTik, Cisco, dual-stack IPv4/IPv6). Telematics Engineering student at Universidad Icesi.

Projects

AromaPacífico Nov 2026 – Present

AWS IoT coffee traceability platform · Team of 5 · GitHub

- Wrote the Terraform roots for dev, test and prod, consuming five versioned module repositories by tag (network, security, IoT ingestion, farm identity, inference) with encrypted, locked S3 state.
- Built plan, deploy, destroy and drift workflows in GitHub Actions with per-environment OIDC roles; deploy applies the saved plan (127 resources in test) and a smoke test traces an IoT reading to DynamoDB in ~10 s.
- Authored the organization's reusable checks: TFLint, Checkov, Gitleaks, actionlint and zizmor at checksum-verified versions, SHA-pinned actions, a tested Python policy checker and a harvest-season deployment freeze gate.

PlazoFiscal Nov 2026

Kubernetes platform for a tax-deadline SaaS · GitHub

- Deployed a five-service application on a three-node k3d cluster that is built and removed with one command, with zero-downtime rolling updates and persistent Redis storage.
- Secured it with default-deny NetworkPolicies and Keycloak MFA through oauth2-proxy in front of every tool and the application; scaled it with KEDA by CPU and by tax-filing dates.
- Built the operations stack: Prometheus and Grafana with Slack alerts, Fluent Bit, Logstash, Elasticsearch and Kibana for logs, Jaeger tracing and Kubecost cost visibility.

KitSalud Sep 2026 – Present

Offline primary-care health kit · Platforms I course · GitHub

- Automated a field-deployable platform (KVM mini PC, MikroTik CCR2004, Wi-Fi AP) with 33 Ansible roles: VLAN segmentation, dual-stack nftables default-deny firewall, Kea DHCPv4/v6, BIND9 and a captive portal.
- Deployed Samba AD as central identity (SMB shares, Grafana and Flask apps over LDAPS), and DHIS2, Kiwix and Jellyfin with Docker Compose behind Caddy and an internal CA, plus encrypted restic backups.
- Centralized metrics and logs (Prometheus, blackbox and node exporters, Grafana, rsyslog) and validated the kit with 13 scripted acceptance tests, working fully offline.

FríoAndes Sep 2026 – Oct 2026

Azure cloud architecture for a cold-chain logistics company · Team of 4 · Website · GitHub

- Designed the hybrid hub-and-spoke network (VPN Gateway, Azure Firewall, DNS Private Resolver, Bastion) with a script-verified IP plan, and the firewall policy table with WAF rate limits for the portal.
- Designed real-time temperature telemetry (IoT Hub, Stream Analytics) with excursion alerts within one minute, the personal-data perimeter, and observability with two SLOs and a Zabbix transition.
- Designed the Microsoft Fabric data platform (incremental ingestion without copying the 1.2 PB archive, AI use cases, cost) and built the project website with Vite and GSAP on GitHub Pages.

Docket Aug 2026 – Sep 2026

GitOps delivery platform on Amazon EKS · Team of 4 · GitHub

- Delivered five polyglot microservices (Go, Spring Boot, Node.js, Python, Vue) through ten connected GitHub Actions pipelines to dev, staging and production on a multi-AZ EKS cluster built from versioned Terraform modules.
- Implemented gated GitOps promotion with ArgoCD and Kustomize: SonarQube and Trivy HIGH/CRITICAL gates, integration tests before staging, recorded approval before production and rollback as a reverted PR.
- Removed long-lived credentials (CI to AWS through OIDC; secrets from SSM through External Secrets and IRSA), routed CloudWatch alarms to Slack through SNS and Lambda, and documented 25 ADRs.

Platform Pipeline Jul 2026

Self-hosted CI/CD, GitOps and observability platform · Personal project · GitHub

- Configured Jenkins entirely as code (JCasC, Job DSL) with tests, a SonarQube quality gate, Gitleaks, Syft SBOM and a Trivy gate; provisioned kind, a local registry and Helm releases with Terraform, using no cloud credentials.

- Delivered through ArgoCD from Gitea with a post-deploy health gate that automatically reverts unhealthy releases; monitored with kube-prometheus-stack and Alertmanager alerts to Slack.

E-commerce on AWS Apr 2026 – May 2026

Full-stack platform with infrastructure as code · Infrastructure III course · GitHub

- Provisioned 10 CloudFormation stacks: VPC across two AZs, EC2 Auto Scaling behind an ALB, RDS PostgreSQL, CloudFront and S3, with CloudWatch, SNS and CloudTrail; Spring Boot API with JWT and a React SPA.

Experience

Universidad Icesi Cali, Colombia

Networking Laboratory Assistant (part-time) Feb 2026 – Present

- Operate the laboratory's internal network: configure layer 2 and layer 3 switches and routers, and troubleshoot connectivity for the classes held in the lab.
- Support faculty and students in lab sessions with device setup and hardware maintenance; selected on merit.

Undergraduate Researcher, Satellite Networks Research Seedbed Jul 2025 – Nov 2025

- Presented a state-of-the-art review of Non-Terrestrial Networks and LEO constellations (Starlink, OneWeb, Telesat Lightspeed, Kuiper) to representatives of the company Celsia.

Teaching Assistant, Algorithms & Programming (part-time) Feb 2024 – Nov 2024

- Supported 30+ students per semester in Java, object-oriented design, data structures and Git.

Technical Skills

- **Cloud:** AWS (EKS, EC2/ASG, VPC, IAM/IRSA, Lambda, API Gateway, IoT Core, DynamoDB, RDS, S3, CloudFront, CloudWatch); Azure architecture (hub and spoke, Azure Firewall, IoT Hub, Fabric); GCP
- **IaC & CI/CD:** Terraform (modules, remote state, drift detection), CloudFormation, Ansible, GitHub Actions (OIDC, reusable workflows), Jenkins (JCasC), ArgoCD, Helm, Kustomize
- **Containers & Kubernetes:** Docker (Compose), Kubernetes (EKS, kubeadm, k3d), KEDA, NetworkPolicies
- **Security & Observability:** SonarQube, Trivy, Gitleaks, Checkov, Syft, External Secrets, Keycloak; Prometheus, Grafana, Alertmanager, ELK, Fluent Bit, Jaeger, Kubecost
- **Networking & Linux:** MikroTik, Cisco, VLAN, IPv4/IPv6, nftables, BIND9, DHCP; Ubuntu Server, KVM
- **Programming:** Python, Bash, Go, Java (Spring Boot), JavaScript/TypeScript, SQL
- **Languages:** Spanish (native), English (B2)

Education

Universidad Icesi Cali, Colombia

B.Eng. in Telematics Engineering, GPA 4.7/5.0 Expected Feb 2028

- Relevant coursework: Infrastructure I–III, Platforms I–II, Infrastructure Automation, Cloud Computing.
- Honor Roll every semester; Honor Scholarship in 2023-2, 2024-2 and 2025-2.
- Capstone (in progress): ORCHID, a platform to deploy and govern AI/ML inference services on Kubernetes in the university's IAsLab, with role-based compute quotas and observability.

Certifications

AWS Academy Graduate: Cloud Operations · Amazon Web Services Jul 2026
AWS Academy Graduate: Cloud Foundations · Amazon Web Services May 2026
Google Cloud Computing Foundations Certificate · Google Cloud Apr 2026
Google Cloud Skill Badges (4) · Google Cloud Mar 2026 – Apr 2026
Cisco DevNet Associate · Cisco Nov 2025
Google Cybersecurity Certificate series · Google 2024 – 2025

Badges verifiable on Credly